Steered Molecular Dynamics for Homomeric Polypeptides

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1 Introduction

2 Methodology

1. For each of the deca-amino acid residues, a probable structure was obtained using AlphaFold server.
2. Each molecular structure was oriented such that its principal (long) axis was aligned along the z-axis.
3. The systems were solvated using CHARMM-GUI. A water box of dimensions $80 \times 40 \times 40$ Å was used, and charge neutralization was performed using KCl ions.
4. Steered Molecular Dynamics (SMD) simulations were carried out using a standardized GROMACS script.
    - **Pulling speeds:** 0.1, 0.01, 0.001
    - **Force field:** CHARMM36m (protein)
    - **Water model:** TIP3P

3 Results

4 Script Availability

All the scripts used are available at the project github.